Pooria Namyar

Senior Researcher, Microsoft Research
pooria.namyar@microsoft.com poorianamyar Pooria Namyar PooriaNamyar

EDUCATION

Doctor of Philosophy in Electrical and Computer Engineering — Aug 2019 – Aug 2025
University of Southern California — Los Angeles, USA
Advisor: Prof. Ramesh Govindan

Master of Science in Computer Science — Aug 2019 – Dec 2023
University of Southern California — Los Angeles, USA

Bachelor of Science in Electrical Engineering — Sep 2014 – July 2019
Sharif University of Technology — Tehran, IRAN

RESEARCH EXPERIENCE

Microsoft Research — Redmond, WA
Senior Researcher — Sep 2025 – Present

Google — Sunnyvale, CA
Research Intern & Student Researcher — Sep 2023 – Jan 2025

Microsoft Research — Redmond, WA
Research Intern & Student Contractor — Jun 2022 – Jan 2023

Microsoft Research — Redmond, WA
Research Intern & Student Contractor — Jun 2021 – Jan 2022

HONORS & AWARDS

- **ACM SIGCOMM Doctoral Dissertation Award** — 2026
- **Google Ph.D. Fellowship in Networking** — 2024 – 2025
- **Selected for NeTS Early Career Investigator Workshop**, *National Science Foundation* — 2025
- **Machine Learning and Systems Rising Star**, *MLCommons* — 2024
- **Ming Hsieh Institute (MHI) Scholarship**, *University of Southern California* — 2023 – 2024
- **4Y Annenberg Graduate Student Fellowship**, *University of Southern California* — 2019 – 2023
- **Top 0.1% in Iranian Nation-wide University Entrance Exam for B.Sc.** — 2014

IMPACT

1. My work on clock synchronization, Firefly [3], is deployed at Google. I developed a mathematical framework to analyze key properties of clock synchronization systems in data centers. For instance, I proved that running distributed clock consensus on a random overlay graph is scalable, converges quickly, and has a near-optimal error. Firefly combines these analyses with system-level optimizations to achieve $\leq$10 ns error.
2. My work, Soroush [8], introduces a general max-min fair allocator that applies to any graph-based resource allocation problem, including traffic engineering and cluster scheduling. Soroush has been deployed in Microsoft and has been managing the traffic across Microsoft's global network since 2023. It has improved the runtime by 3$\times$ on average (up to 5.4$\times$) without compromising fairness and efficiency. In this work, I designed novel algorithms with provable guarantees to solve multi-path max-min fair allocation using a single fast optimization – achieving this *for the first time* in the networking literature.
3. My work, MetaOpt [9], opens up a new area of research on scalable, general, and user-friendly performance analyzers. MetaOpt enables practitioners to identify and fix the pathological behavior of their algorithms before deployment. It has attracted interest from both academia and industry. For instance, I collaborated with researchers at ETH Zürich to analyze the trade-offs in a new packet scheduling heuristic using MetaOpt [7]. MetaOpt has also identified and fixed inefficiencies in key production heuristics at Microsoft.

PUBLICATIONS

[Refereed Conference Publications]

1. Near-optimal Online Traffic Engineering
   Arvin Ghavidel, **Pooria Namyar**, Nikolai Matni, Walter Willinger, Ramesh Govindan
   ACM SIGCOMM 2026
2. Heuristic Analysis from Source Code via Symbolic-Guided Optimization (link)
   Pantea Karimi, Siva Kakarla, Santiago Segarra, Ryan Beckett, **Pooria Namyar**, Mohammad Alizadeh, Behnaz Arzani
   NSDI 2026
3. Firefly: Scalable, Ultra-Accurate Clock Synchronization for Datacenters (link)
   Pooria Namyar, Yuliang Li, Weitao Wang, Nandita Dukkipati, KK Yap, Junzhi Gong, Chen Chen, Peixuan Gao, Devdeep Ray, Gautam Kumar, Yidan Ma, Ramesh Govindan, Amin Vahdat
   ACM SIGCOMM 2025
4. ZENITH: Towards A Formally Verified Highly-Available Control Plane (link)
   Pooria Namyar, Arvin Ghavidel, Mingyang Zhang, Harsha V. Madhyastha, Srivatsan Ravi, Chao Wang, Ramesh Govindan
   ACM SIGCOMM 2025
5. Raha: A General Tool to Analyze WAN Degradation (link)
   Behnaz Arzani, Sina Taheri, **Pooria Namyar**, Ryan Beckett, Siva Kakarla, Elnaz Jalilipour
   ACM SIGCOMM 2025
6. Enhancing Network Failure Mitigation with Performance-Aware Ranking (link)
   Pooria Namyar, Arvin Ghavidel, Daniel Crankshaw, Daniel Berger, Kevin Hsieh, Srikanth Kandula, Ramesh Govindan, Behnaz Arzani
   NSDI 2025
7. Everything Matters in Programmable Packet Scheduling (link)
   Albert Gran Alcoz, Balázs Vass, **Pooria Namyar**, Behnaz Arzani, Gábor Rétvári, Laurent Vanbever
   NSDI 2025
8. Solving Max-Min Fair Resource Allocations Quickly on Large Graphs (link)
   Pooria Namyar, Behnaz Arzani, Srikanth Kandula, Santiago Segarra, Daniel Crankshaw, Umesh Krishnaswamy, Ramesh Govindan, Himanshu Raj
   NSDI 2024
9. Finding Adversarial Inputs for Heuristics using Multi-level Optimization (link)
   Pooria Namyar, Behnaz Arzani, Ryan Beckett, Santiago Segarra, Himanshu Raj, Umesh Krishnaswamy, Ramesh Govindan, Srikanth Kandula
   NSDI 2024
10. Optimal Oblivious Routing for Structured Networks (link)
    Sucha Supittayapornpong, **Pooria Namyar**, Mingyang Zhang, Minlan Yu, Ramesh Govindan
    IEEE INFOCOM 2022
11. A Throughput-Centric View of the Performance of Datacenter Topologies (link)
    Pooria Namyar, Sucha Supittayapornpong, Mingyang Zhang, Minlan Yu, Ramesh Govindan
    ACM SIGCOMM 2021

[Refereed Workshop Publications]

12. End-to-End Performance Analysis of Learning-enabled Systems (link)
    Pooria Namyar, Michael Schapira, Ramesh Govindan, Santiago Segarra, Ryan Beckett, Siva Kakarla, Behnaz Arzani
    ACM HotNets 2024
13. Towards Safer Heuristics With Xplain (link)
    Pantea Karimi, Solal Pirelli, Siva Kakarla, Ryan Beckett, Santiago Segarra, Beibin Li, **Pooria Namyar**, Behnaz Arzani

ACM HotNets 2024

14. Minding the gap between Fast Heuristics and their Optimal Counterparts (link)
    Pooria Namyar, Behnaz Arzani, Ryan Beckett, Santiago Segarra, Himanshu Raj, Srikanth Kandula
    ACM HotNets 2022

[Refereed Journal Publications]

15. Optimal Oblivious Routing with Concave Objectives for Structured Networks (link)
    Kanatip Chitavisutthivong, Sucha Supittayapornpong, **Pooria Namyar**, Mingyang Zhang, Minlan Yu, Ramesh Govindan
    IEEE/ACM Transactions on Networking 2023

[Under Submission/Preparation]

16. Switchcraft: AI Model Router for Agentic Tool Calling
    Sharad Agarwal, **Pooria Namyar**, Alec Wolman, Rahul Ambavat, Ankur Gupta, Qizheng Zhang
17. A Performance Analyzer for a Public Cloud's ML-Augmented VM Allocator
    Roozbeh Bostandoust, **Pooria Namyar**, Siva Kesava Reddy Kakarla, Ryan Beckett, Santiago Segarra, Eli Cortez, Ankur Mallick, Kevin Hsieh, Rodrigo Fonseca, Mohammad Hajiesmaili, Behnaz Arzani
18. Explanation-Guided Heuristic Design with LLMs
    Pantea Karimi, Dany Rouhana, **Pooria Namyar**, Siva Kesava Reddy Kakarla, Venkat Arun, Behnaz Arzani

PATENTS

1. Flow Network Intermediate Representation For Optimization Problems, US patent application, Published 6/2025
2. Solving Max-min Fair Resource Allocation at Large Scale, US patent application, Published 9/2024
3. Network Traffic Control Using Estimated Maximum Gap, US patent application, Published 3/2024
4. Impact-Aware Mitigation For Computer Networks, US patent application, Published 3/2023
5. Artificial Intelligence Analysis For Holistic System Design, US patent application, under review
6. Network Performance Degradation Analyses And Remediation, US patent application, under review
7. Explanation-guided Heuristic Design, US patent application, under review

INVITED TALKS & PRESENTATIONS

Toward Reliable and Scalable Network Algorithms

Microsoft Research, Redmond, WA, USA — Mar 2025
University of Maryland, College Park, MD, USA — Mar 2025

Heuristic Analyzers for Networked Systems

ACM HotNets 2024, Irvine, CA, USA (Ph.D. short talk) — Nov 2024

End-to-End Performance Analysis of Learning-enabled Systems

ACM HotNets 2024, Irvine, CA, USA — Nov 2024
Microsoft Research, Virtual — Nov 2024

Finding Adversarial Inputs for Heuristics using Multi-level Optimization

Rising Stars workshop at NVIDIA, CA, USA — Jul 2024
Google Systems Talk, Sunnyvale, CA, USA (Host: Jeff Mogul) — Jun 2024
NSDI 2024, Santa Clara, CA, USA — Apr 2024
Microsoft Research, Virtual — Apr 2024

Solving Max-Min Fair Resource Allocation at Large Scale

NSDI 2024, Santa Clara, CA, USA — Apr 2024
Microsoft Research, Virtual — Sep 2022

Minding the gap between Fast Heuristics and their Optimal Counterparts

ACM HotNets 2022, Austin, Texas, USA — Nov 2022

Microsoft Research, Virtual — Sep 2022

A Throughput-Centric View of the Performance of Datacenter Topologies

Google Networking Research Summit, Virtual — Mar 2022
11th Annual Research Festival, University of Southern California, USA — Oct 2021
Microsoft Research, Virtual — Aug 2021
ACM SIGCOMM 2021 Conference, Virtual — Aug 2021

TEACHING & MENTORING

Teaching

- Teaching Assistant, Introduction to Internetworking (CSCI 353), Fall 2024, University of Southern California
- Guest Lecturer, Networked Systems in Cloud Computing (CS 599/656), Spring 2024, University of Southern California
- Teaching Assistant, Networked Systems in Cloud Computing (CS 599/656), Spring 2024, University of Southern California
- Teaching Assistant, Data Networks, Fall 2018, Sharif University of Technology
- Teaching Assistant, Introduction to Machine Learning, Fall 2018, Sharif University of Technology
- Teaching Assistant, EE Principles Laboratory, Fall 2016, Sharif University of Technology

Mentoring

- Arvin Ghavidel (PhD), University of Southern California — Fall 2022 – Present
- Roozbeh Bostandoost (PhD), University of Massachusetts Amherst — Spring 2024 – Present
- Pantea Karimi (PhD), Massachusetts Institute of Technology — Fall 2024 – Present
- Neal Sharma (BS), University of California, Berkeley — Fall 2022 – Summer 2023
- Smit Kadvani (MS), University of Southern California — Fall 2021

PROFESSIONAL SERVICE

Service

- Program Committee: ACM SIGCOMM 2026, ATC 2026, CoNEXT 2026, APNet 2026
- Pre-Review Task Force: NSDI 2025
- AE Reviewer: ACM SIGCOMM 2024, ACM SOSP 2023, ACM SIGCOMM 2023
- Reviewer: IEEE SPAWC 2024
- External Reviewer: NSDI 2022, ESA 2025

Outreach

- Viterbi Graduate Mentor, University of Southern California, 2024 – 2025
- Volunteer in CS@SC Coding Camps, Viterbi K-12 STEM Center, 2024
- Panelist, Sharing PhD journey with Undergraduate Students, University of Southern California, 2024
- Womxn in Engineering (WIE) mentor, University of Southern California, 2023 – 2024
- Organizer, MHI Annual Research Festival, 2023

MEDIA COVERAGE

- Urs Hölzle's post (Oct 23, 2025) about our work on clock synchronization.
- Viterbi School of Engineering (Oct 3, 2024), Ph.D. Candidate Pooria Namyar Among 41 Named MLCommons Rising Stars.
- Viterbi School of Engineering (Jul 10, 2024), Student Builds an 'Angel' to Lift Up Microsoft's Wide-Area Network.
- Microsoft Research Blog (Jun 20, 2024), Microsoft's official announcement about the deployment of Soroush.
- Microsoft Research Blog (Jan 23, 2024), MetaOpt: Examining, explaining, and improving heuristic performance.